

Agent Execution & Environment Rules

Project Architecture (Monorepo Structure) The agent must strictly adhere to this folder structure and never invent new root directories:

- `/apps/upaj/` - Expo React Native codebase for the Farmer app.
- `/apps/mandi/` - Expo React Native codebase for the Buyer app.
- `/backend/` - Python FastAPI codebase.
- `/shared/` - Shared UI components, API clients, and TypeScript interfaces.
- `/hardware/` - C++ / Arduino `.ino` files for the ESP32.

Environment Variables (`.env`) The agent must create a `.env.example` file in the backend with the following expected keys. It must NEVER commit real secret keys to the codebase:

- `DATABASE_URL` (PostgreSQL connection string)
- `WEATHER_API_KEY` (OpenWeatherMap API key)
- `AGMARKNET_API_KEY` (Government Mandi API integration)
- `JWT_SECRET` (For mobile authentication tokens)

Execution & Testing Commands The agent must use these exact commands when verifying its work:

- **Backend:** Navigate to `/backend` and run `uvicorn main:app --reload`
- **Mobile Apps:** Navigate to `/apps/upaj` or `/apps/mandi` and run `npx expo start`
- **Database:** Run `alembic upgrade head` for schema migrations.

CI/CD Pipeline (GitHub Actions) The agent must generate a GitHub Actions workflow (`.github/workflows/deploy.yml`) to enforce automated testing and deployment.

- **Trigger:** Execute pipeline automatically on every push to the `main` branch.
- **Test Gate:** Check out code, set up Python, install dependencies, and execute `pytest`. If any tests fail, the pipeline must block the deployment process entirely.
- **Deploy Gate:** Only trigger the production deployment workflow if the test suite passes successfully.

Production Deployment Paths The agent must configure the backend infrastructure to support one of the following automated cloud deployment strategies:

- **Option A: Render Deployment:** The agent must configure the FastAPI service with the correct build command (`pip install -r requirements.txt`) and start command (`uvicorn main:app --host 0.0.0.0 --port $PORT`). The `DATABASE_URL` must be injected via the Render dashboard to connect to the managed PostgreSQL instance.

- **Option B: AWS Deployment (App Runner / ECS):** The agent must write a production-ready `Dockerfile` for the backend. The CI/CD pipeline must securely authenticate with AWS using OIDC or IAM secrets, build the Docker image, push it to Amazon Elastic Container Registry (ECR), and trigger an automatic deployment rollout to either AWS App Runner or ECS Fargate.

Seed Data Requirement Before executing any tests on the Hyperlocal Commerce Engine, the agent must generate a Python seeding script (`seed_mock_data.py`) that populates the PostgreSQL database with at least 10 dummy farmers, 5 dummy buyers, and 20 active crop listings using mock PostGIS coordinates centered around Kolar, Karnataka.